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Efficient perovskite solar modules with an ultra-long processing window enabled by cooling stabilized intermediate phases†

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Perovskite solar cells (PSCs) have shown promising progress in efficiency and stability, but their application needs further development from small-area cells to large-area modules. When fabricating solar cell modules on large-area substrates, it takes a longer time to deposit and process the thin film than for small area devices. Therefore, it is required to expand the processing window of the solution process. Here, we showed that combining *N*-methyl-2-pyrrolidone solvent with a cooling strategy can generate more stable FA-based perovskite intermediates in two-step deposition, realizing a longer annealing window. The power conversion efficiency (PCE) of 25.21% for small-area devices (0.045 cm²) and 23.60% for large-area devices (1.00 cm²) was achieved. It is found that cooling the intermediate phase at a temperature close to 0 °C can suppress the formation of δ -phase FAPbI₃ and expand the annealing window by 20-fold (from 9 min to 180 min). With these cooling strategies, we have successfully fabricated uniform perovskite films in a large area of 45 cm². PSC mini-modules made by this method achieved state-of-the-art efficiencies of 22.34% and a certified efficiency of 21.51%. More importantly, even with an annealing delay time of 180 min, the modules attain a decent PCE of 20.89%, showing an ultra-long processing window for fabricating efficient PSCs. Our strategy of stabilizing the perovskite intermediate phase brings great flexibility to the large-scale production of perovskite solar modules.

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Broader context

One key challenge of commercializing perovskite solar cells (PSCs) is the deposition of high-quality perovskite films in a large area. Nevertheless, deposition of perovskite films through solution processes usually forms intermediates that are easily affected by the environment. In addition, the intermediates are unstable and needed to be converted to the perovskite phase within a short time window of seconds to a few minutes. The sensitivity and short processing window of perovskite intermediates hinders the deposition of pure phase FAPbI₃ films in a large area. In this context, we showed that combining *N*-methyl-2-pyrrolidone solvent with an intermediate cooling strategy in two-step deposition can achieve more stable FA-based perovskite intermediates, realizing a longer annealing window of up to 2 hours. Our strategy of stabilizing the perovskite intermediate phase and achieving an ultra-long processing window can facilitate up-scaling deposition of perovskite films on a larger substrate area, and brings great flexibility to module fabrication, which will pave the way towards high-performance perovskite modules.

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Introduction

In the past decade, hybrid organic–inorganic metal halide perovskite solar cells (PSCs) have received considerable attention as next generation photovoltaic technology due to their high efficiency,^{1,2} tuneable bandgaps and easy fabrication.^{3–6} Power conversion efficiencies (PCEs) of up to 26.1% have been achieved in laboratory-based small-area PSCs ($<0.1\text{ cm}^2$),⁷ catching up with the well-developed Si solar cells. In addition, PSCs have lower material costs and can be fabricated over large areas by solution methods, making them promising for future commercialization.^{8,9} However, scaling from laboratory-scale small-area cells to industrial-scale perovskite solar modules (PSMs) ($>10\text{ cm}^2$) while retaining high PCE has posed serious challenges.

The most important prerequisite for module fabrication is the uniform deposition of a high-quality large-area perovskite film. The nucleation and crystal growth of a perovskite in a solution are largely uncontrollable. Strategies to avoid non-uniform crystal growth in the solution process include solvent quenching, vacuum-flash, hot coating and two-step deposition.^{10–14} Such methods usually form perovskite intermediate phases that still contain solvent molecules. However, the intermediate phases are unstable and easily affected by deposition conditions, such as humidity and temperature.^{15,16} Conversion of the intermediate phase must be performed within a short period of time in seconds to minutes. The intermediate phase can easily decompose at room temperature. Delaying the processing of the intermediate phase will ultimately result in low-quality films^{17,18} and poor device performance. The larger the device area is, the more difficult achieving a uniform film is, due to the considerable time needed to deposit on a large size substrate. For meter scale commercial solar modules, the deposition and handling time of each substrate could take minutes to tens of minutes depending on the process machinery.^{19,20} During the long film deposition process, the initial and last part of the deposited film experience largely different delay times upon the annealing step, which inevitably leads to non-uniformity in the final perovskite films. A large size substrate also increases the time for carrying and handling of the intermediate film. Therefore, achieving a stable perovskite intermediate phase has become a key challenge for industrial module fabrication.

To stabilize the perovskite intermediate phase and slow down the nucleation and crystal growth, researchers have proposed strategies such as adding specific solvents or additives, which have shown some success in preparing large-area perovskite films. Zhu *et al.*,⁶ by adding high boiling point solvent *N*-methyl-2-pyrrolidone (NMP) and methylammonium chloride (MACl) to the perovskite precursor to form a stable intermediate phase, extended the time window for the delay before annealing from a few seconds up to 8 min, achieving 13.3% efficiency for a methylammonium lead iodide (MAPbI₃)-based mini-module (12.6 cm^2). However, shortcomings in MA-based PSCs have gradually been revealed, and the stability of these materials is unsatisfactory under high temperature or light conditions.

Formamidinium (FA)-based perovskites show good thermal and light stability. In addition, the narrower optical bandgap of FAPbI₃ (1.48 eV) is closer to the ideal solar cell band gap predicted by the Shockley–Queisser limit.^{20,21} Thus, researchers are increasingly focusing on the preparation of high-quality FA-based perovskite films. Unfortunately, the crystal growth and phase control of FA-based perovskites are more difficult to manage than those of MA-based perovskites, especially in two-step deposition. At room temperature (RT), the intermediate easily transforms into non-photoactive δ -phase FAPbI₃, which undermines its photovoltaic performance.²² Chen *et al.*²³ added the nonvolatile additive diphenyl sulfoxide (DPSO) in FAC perovskite precursors to form more stable intermediate phases and increased the nucleation barrier. The wet precursor film is stable for up to 8 min, achieving an efficiency of 16.6% in a mini-module. Yan *et al.*²⁴ reported a perovskite precursor solution consisting of 2-methoxyethanol (2-Me) and 1,3-dimethyl-imidazolidinone (DMI), which stabilized the intermediate phase to enable large-scale production of PSMs. However, the above additives and solvents still show limitations in maintaining a stable intermediate phase for more than 10 min, which limits the preparation of PSMs in a larger area. Perovskite nucleation is closely tied to the critical Gibbs free energy (ΔG^*), primarily influenced by solute particle size, solution saturation, and temperature.²⁵ Lowering the temperature of the precursor film favorably controls the nucleation process and film growth. Chen *et al.* show that by lowering the storage temperature of the perovskite intermediate films from 70 °C to 30 °C, the nucleate of the δ -phase can be suppressed and more α -phase crystals are seeded in the intermediate films, which extends the processing window of the intermediate phase.²⁶

In this work, we report a combined solvent engineering and substrate cooling strategy to greatly extend the processing time window of the FAPbI₃ perovskite intermediate from a few minutes to tens of minutes. The strategy of stabilizing the perovskite intermediate phase at low temperatures (LT) was successfully applied in the fabrication of large-area FAPbI₃-based PSMs. DMF/NMP is used to replace conventional solvent DMF/DMSO for PbI₂ deposition. The stronger binding energy between NMP and PbI₂ leads to a more stable PbI₂ adduct and MA_xFA_(1-x)PbI_(3-x)Cl_x intermediate, therefore extending the annealing delay time during two-step deposition. Cooling the MA_xFA_(1-x)PbI_(3-x)Cl_x intermediate to a temperature close to 0 °C can further extend the process time up to 180 min, a remarkably long processing window. The ultra-long stable time for the perovskite intermediate films adds much flexibility to the two-step fabrication process and greatly facilitates large-area film deposition. Based on this method, a state-of-the-art PCE of 25.21% for small-area devices (0.045 cm^2) and 23.60% for large-area devices (1.00 cm^2) was obtained. The corresponding $5\text{ cm} \times 5\text{ cm}$ and $6.5\text{ cm} \times 7\text{ cm}$ PSMs demonstrate high efficiencies of up to 22.34% and 21.91% with an aperture area of 16.8 cm^2 and 27.81 cm^2 . The encapsulated PSMs under ISOS-O-1 conditions exhibit a T_{97} lifetime of more than 1000 hours. More importantly, even with an ultra-long annealing delay time of 180 min, the PSM can still obtain a PCE of

20.89%, much longer than the 5 min maximum delay time with the conventional method. Our findings reveal the importance of stabilizing perovskite intermediates for achieving large-area uniform perovskite films and efficient PSMs.

Results and discussion

Fig. 1 presents the typical procedures for preparing FA-based perovskite films using two-step deposition. Typically, there are three steps involved in the two-step deposition: (1) deposition of PbI_2 , in which a PbI_2 -solvent adduct wet film is obtained after annealing at a moderate temperature ($\sim 70^\circ\text{C}$); (2) coating of formamidinium ammonium salts onto the PbI_2 -solvent film to form perovskite-solvent intermediate phases; and (3) thermal annealing of the wet film at a temperature of $\sim 150^\circ\text{C}$ and a humidity of 30–40% to convert the intermediate phase into FAPbI_3 perovskite. Considering the different solvents and atmospheres used in different steps, the procedure is usually segmented into steps with an unavoidable processing delay at each step. Processing time t_1 for the PbI_2 adduct films and t_2 for the perovskite intermediate phase films are inevitable in these procedures. The processing time t_2 in the third step is particularly important for achieving high performance devices and is typically less than 5 min, otherwise the intermediate phase would decompose leading to pin-holes and δ -phase in the final FAPbI_3 perovskite films. Depositing a large-area film requires a longer time for each coating, storage and transfer step. Therefore, large processing windows for both t_1 and t_2 are highly desired to accommodate the long delay times necessary in large-scale PSM fabrication.

Previous works have shown that the process window is increased in one-step deposition by NMP,^{6,20,23} which has a higher boiling point than DMSO (Table S1, ESI†). To explore the roles of NMP and the corresponding changes in the process window in the two-step deposition, we used the DMF/NMP (D/N) solvent system instead of DMF/DMSO (D/D) for depositing PbI_2 in Step 1. The crystallization of the PbI_2 in D/D and D/N solvents was characterized by optical microscopy. In D/D solvent, needle-like PbI_2 -DMSO crystals initially formed from the edge (Fig. S1a and S2a, ESI†).²⁴ Subsequently, thicker dendrite

structure PbI_2 -2DMSO crystals appeared. The formation of these PbI_2 adducts with DMSO could suppress the conversion of PbI_2 into perovskite.^{25,27,28} As a comparison, only lamellar PbI_2 -NMP crystals appeared in D/N solvent (Fig. S1b and S2b, ESI†), and the crystallization rate was much slower. The SEM image (Fig. S3a, ESI†) also shows that the PbI_2 @D/D film was smooth and compact, while the PbI_2 @D/N film presented a laminar porous structure (Fig. S3b, ESI†), which provided channels for formamidinium halide penetration and reaction.²⁹ The different solvent adducts and morphology of the PbI_2 formed in the two solvent systems would influence the PbI_2 conversion to perovskite (Fig. S3c and d, ESI†).

We then turn our attention to the processing time t_2 . Most previous studies used a short t_2 , usually only a few tens of seconds to several minutes.^{6,30,31} FAI solution coating takes more time for large-area deposition, thus requiring a longer processing window for t_2 . To determine the difference in the conversion process with the two solvent systems, we compared photographs of the PbI_2 films after FAI coating with different t_2 values (Fig. S4, ESI†). The intermediates obtained with the D/D solvent appeared red and the surface became significantly cloudy after 5 min. In contrast, the intermediates obtained with the D/N solvent showed only slight changes with increasing t_2 , the colour changed from light tan to tan, but the film remained clear, which indicates that the obtained intermediates are more stable.

The crystallization of FAPbI_3 in the second step was also observed by optical microscopy. With the D/D solvent, butterfly-shaped FAPbI_3 crystals and large hexagonal δ -phase crystals (Fig. S5a, ESI†) appeared immediately after FAI deposition.²⁰ Some needle-shaped crystals gradually appeared after 1 min and dominated the whole film after 5 min, obviously suppressing the growth of the FAPbI_3 crystals. With the D/N solvent, the crystal growth was much slower, with only part of the film crystallizing after 5 min, and only butterfly-shaped FAPbI_3 crystals were observed (Fig. S5b, ESI†). The slower crystallization rate and singular morphology suggest a longer process time t_2 and purer phase from the D/N solvent system.

To reveal the mechanism of longer process time from the D/N solvent system, Fourier transform infrared (FTIR) and nuclear magnetic resonance (^1H NMR) spectroscopy were used

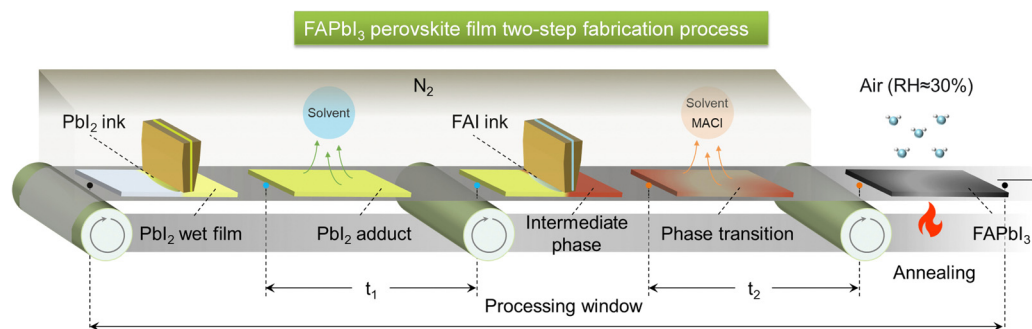


Fig. 1 Schematics of the two-step fabrication process of FAPbI_3 perovskite films. The background colour indicates a different atmosphere at different steps. The processing windows for PbI_2 and perovskite intermediates are shown as t_1 and t_2 , respectively.

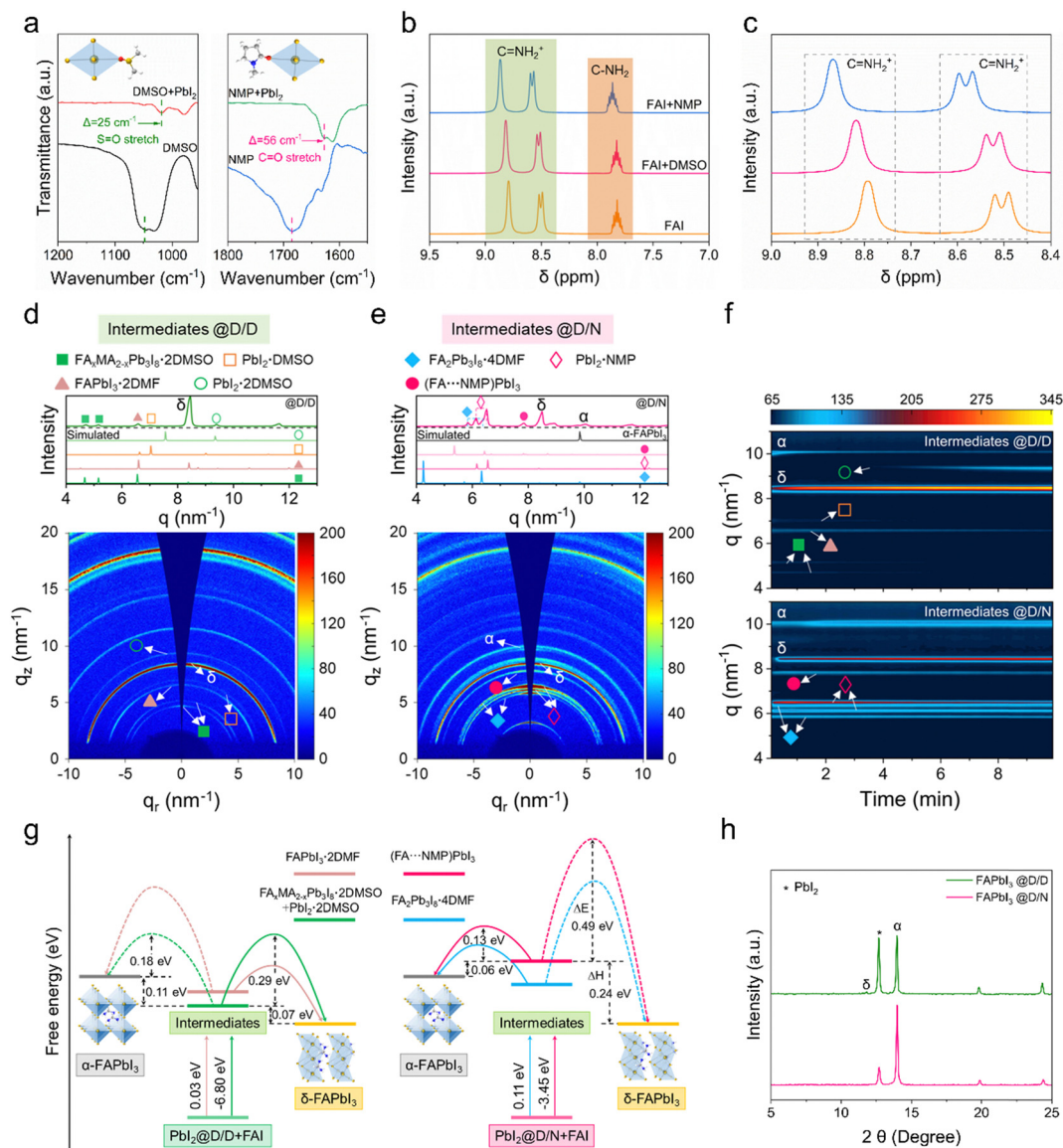


Fig. 2 Interactions between solvent molecules and perovskite precursors and their effects on the nucleation and crystallization of FA-based perovskites. (a) Representative FTIR peaks of DMF, NMP and their PbI_2 mixtures. (b) and (c) ^1H NMR spectra of FAI, FAI + DMSO and FAI + NMP. (d) and (e) 2D GIWAXS images and the integral spectra of the intermediate@D/D (d) and intermediate@D/N perovskite intermediate films (e). The incidence angle is 1° . q_z , out-of-plane scattering vector; q_r , in-plane scattering vector. (f) *In situ* GIWAXS pattern of the intermediate@D/D and intermediate@D/N film as a function of time. (g) Free-energy diagram of the intermediates, α -FAPbI₃ and δ -FAPbI₃ perovskite formed in different solvent systems. (h) The XRD patterns of the annealed perovskite films FAPbI₃@D/D and FAPbI₃@D/N from different solvent systems.

to compare the interactions between the solvents and the perovskite precursors. The S=O of DMSO showed a shift of 25 cm^{-1} , while C=O of NMP showed a larger shift of 56 cm^{-1} , demonstrating a stronger binding between PbI_2 and NMP (Fig. 2a and Fig. S6a, ESI[†]).³² The strong interaction between PbI_2 and NMP explained the slower crystallization of PbI_2 and longer process time t_1 in step 1.

Later on, FAI solution was dropped onto the PbI_2 films to form intermediates in step 2 of the process. The intermediates are denoted as intermediate@DMSO and intermediate@NMP for different solvents. The FTIR peak corresponding to the C=N of FAI is slightly shifted in intermediate @DMSO,

as shown in Fig. 2a, while the peak shows a larger shift from 1730 cm^{-1} to 1712 cm^{-1} in intermediate@NMP, suggesting a stronger interaction between FAI and NMP. The FTIR peak corresponding to the N-H of FA^+ also showed a clear shift in the intermediate@NMP samples (Fig. S6b, ESI[†]), also indicating the formation of hydrogen bonds between NMP and FA^+ .³³ Nuclear magnetic resonance (^1H NMR) spectra showed that the amino proton peak of FAI at $\delta = 8.79 \text{ ppm}$ shifted to a higher field after mixing with NMP, compared to DMSO (Fig. 2b and c). This indicated stronger hydrogen bonding interactions between NMP and FA^+ than between DMSO and FA^+ . The strong FAI-NMP interaction led to more stable perovskite intermediates, and

achieved a longer t_2 of up to 9 min (Fig. S4, ESI†). In contrast, the perovskite intermediate obtained with the D/D solvent was unstable and readily converted into an opaque film at the same time.

FAPbI₃ has two phases at RT. One is the photovoltaic-active α -phase and the other is the non-photoactive δ -phase. It is important to understand and control the phase transition of the perovskite intermediate at step 2 to achieve high performance PSCs. The phase transition pathway of perovskite intermediates was investigated using *in situ* UV-vis absorption. As shown in Fig. S7a (ESI†), when t_2 increases to 20 s, the absorption edge of intermediate@D/D quickly shifted from 700–800 nm to 300–400 nm, resembling a typical δ -FAPbI₃ absorption.³⁴ The absorption spectra of the intermediate@D/N film showed almost no change with t_2 up to 65 s, and changed at a slower rate thereafter. The absorption edge (~ 550 nm) did not match both the α -phase and δ -phase of FAPbI₃, indicating that intermediate@D/N forms a new and relatively stable phase combining FAPbI₃ and NMP. The average absorbance (450–800 nm) of the intermediate films as a function of t_2 is presented in Fig. S7b (ESI†), which also showed the different stability of intermediate@D/D and intermediate@D/N.

We further used synchrotron-based grazing-incidence wide-angle X-ray scattering (GIWAXS) and XRD to analyse the intermediate phase evolution. The 2D GIWAXS patterns and their one-dimensional (1D) integral spectra of the intermediate@D/D film showed a strong peak (scattering vector $q = 8.44 \text{ nm}^{-1}$) corresponding to the strong δ -FAPbI₃ phase and several small peaks corresponding to multiple solvent-containing intermediate phases FA_xMA_{2-x}Pb₃I₈·2DMSO ($q = 4.72 \text{ nm}^{-1}$ and $q = 5.16 \text{ nm}^{-1}$), FAPbI₃·2DMF ($q = 6.63 \text{ nm}^{-1}$), PbI₂·DMSO ($q = 7.04 \text{ nm}^{-1}$) and PbI₂·2DMSO ($q = 9.38 \text{ nm}^{-1}$) (Fig. 2d). However, almost no diffraction peak corresponding to the α -FAPbI₃ phase can be observed at $t_2 = 12$ min, suggesting that the intermediate@D/D film tended to convert into the δ -FAPbI₃ phase over time. The intermediate@D/N film (Fig. 2e) showed a strong PbI₂·NMP peak ($q = 6.15 \text{ nm}^{-1}$ and $q = 6.50 \text{ nm}^{-1}$), and an α -FAPbI₃ peak ($q = 10.03 \text{ nm}^{-1}$), together with a tiny δ -FAPbI₃ peak. Solvent containing intermediate phases were also presented as FA₂Pb₃I₈·4DMF ($q = 5.83 \text{ nm}^{-1}$ and $q = 6.38 \text{ nm}^{-1}$) and (FA·NMP)PbI₃ ($q = 7.85 \text{ nm}^{-1}$).

The phase evolution of the perovskite intermediate with increasing t_2 was monitored by *in situ* GIWAXS. As shown in Fig. 2f, at the initial state, the intermediate@D/D film contains a variety of intermediate phases, the strength of the δ -phase is relatively weak, and an α -phase appears. With the extension of t_2 , the δ -phase gradually dominates the film. However, the initial intermediate@D/N film is dominated by PbI₂·NMP, and the strength of the δ -phase is relatively weak. With t_2 increasing, the PbI₂·NMP and (FA·NMP)PbI₃ phase gradually weaken, and the α -phase increases, while the intensity of the δ -phase remains constant, indicating that the PbI₂·NMP and (FA·NMP)PbI₃ phase tend to directly transform into the α -phase, thereby reducing the ratio of δ -phase in the final films. The changes of α -phase and δ -phase over t_2 in the two intermediate phase films are shown in Fig. S11 (ESI†). The intensity

changes of other intermediate phases can be seen in Fig. S8–S10 (ESI†). The intensity of FA_xMA_{2-x}Pb₃I₈·2DMSO, FAPbI₃·2DMF and PbI₂·DMSO in the intermediate@D/D film is relatively low, while the intensity of the δ -phase is higher, mainly due to their rapid transformation into δ -phase. However, the intensity of PbI₂·NMP and (FA·NMP)PbI₃ in the intermediate@D/N film are relatively higher, and gradually decrease with t_2 due to conversion into α -phase. Even if t_2 is delayed for 9 min, PbI₂·NMP, (FA·NMP)PbI₃ and α -phase co-exist, which indicates that the intermediate@D/N film has better stability.

The different phase transition pathway is mainly a result of different free energy and transition energy barriers of the intermediates. The free energy and phase transition energy barrier (Table S2, ESI†) of the intermediate phases with different solvents were calculated using density functional theory (DFT). The preferred crystal structures of possible solvent-perovskite adducts are shown in Fig. S12 (ESI†), which are used to generate the energy diagram in Fig. 2g. For the D/D system, the dominate intermediate phase is FA_xMA_{2-x}Pb₃I₈·2DMSO (green) and FAPbI₃·2DMF (brown) according to the GIWAXS spectra of the intermediate@D/D film (Fig. 2d). Their energy barrier towards δ -FAPbI₃ is equal to or lower than α -FAPbI₃, resulting in preferred transition to δ -FAPbI₃. The GIWAXS integral spectra of the intermediate@D/N film (Fig. 2e) showed PbI₂·NMP, (FA·NMP)PbI₃ and FA₂Pb₃I₈·4DMF peaks. (FA·NMP)PbI₃ and FA₂Pb₃I₈·4DMF both tend to convert to α -FAPbI₃ due to a lower transition barrier towards α -FAPbI₃ than δ -FAPbI₃. This is consistent with our results above.

Then the intermediate films were annealed at 150 °C to gain the final perovskite film. The FAPbI₃@D/D film derived from annealing the intermediate@D/D film at 150 °C contains α -phase, but with a large amount of PbI₂ and δ -phase (Fig. 2h). The FAPbI₃@D/N film showed a dominant α -phase with only a tiny amount of PbI₂. Furthermore, we compared the crystal quality of perovskite films transformed from intermediate phase films with different storage times. In the D/D system, the crystal quality deteriorated and residual PbI₂ increased with increasing t_2 according to the XRD pattern (Fig. S14a, ESI†). When the processing time t_2 was greater than 5 min, the perovskite film showed an obvious colour change, indicating the occurrence of an impurity phase (Fig. S13, ESI†). The perovskite films from the D/N system showed good crystallinity and only a small amount of PbI₂ even with a high t_2 value of 9 min (Fig. S14b, ESI†). Therefore, FAPbI₃@D/D could only form a film with high crystal quality within $t_2 = 5$ min, while FAPbI₃@D/N accommodated up to 9 min, which was in good agreement with the SEM results (Fig. S15, ESI†) of the FAPbI₃@D/D and FAPbI₃@D/N films with different t_2 values.

Although the use of the NMP solvent extended the processing window for t_2 in the two-step deposition, the intermediate@D/N phase still changed with t_2 longer than 10 min. The complex and unstable intermediate phase substantially affected the final film quality. Further stabilizing the intermediate phase is desirable for scalable fabrication of large-area perovskite films.

At RT, the intermediate phase is unstable and decomposes with time delay, and the higher the temperature, the faster

the change rate (Fig. S16, ESI†). Therefore, we proposed a new strategy of cooling the perovskite intermediate to LT (0–5 °C) using a cooling stage to suppress the decomposition of the intermediate phase (Fig. S17, ESI†). Unlike the transformation at RT, the intermediate@D/D film at LT did not turn yellow and maintained a smooth surface even after 180 min. The intermediate@D/N film stored at 0 °C showed only a slight colour change in the same period. Both of the intermediate phase films turned black after annealing at 70 °C

for 5 min, indicating that the α -FAPbI₃ phase was formed in the films.

The phase transition of the intermediate at 0 °C was characterized using *in situ* PL spectroscopy. Unlike the rapid quenching of the PL intensity in the intermediate@D/D film at RT (Fig. 3a), the PL intensity of the intermediate@D/D film at LT increases and gradually redshifts with storage time t_2 up to 60 min (Fig. 3c and Fig. S18a, ESI†). This process shows that the intermediate phase can remain stable for a longer time at LT.³⁵

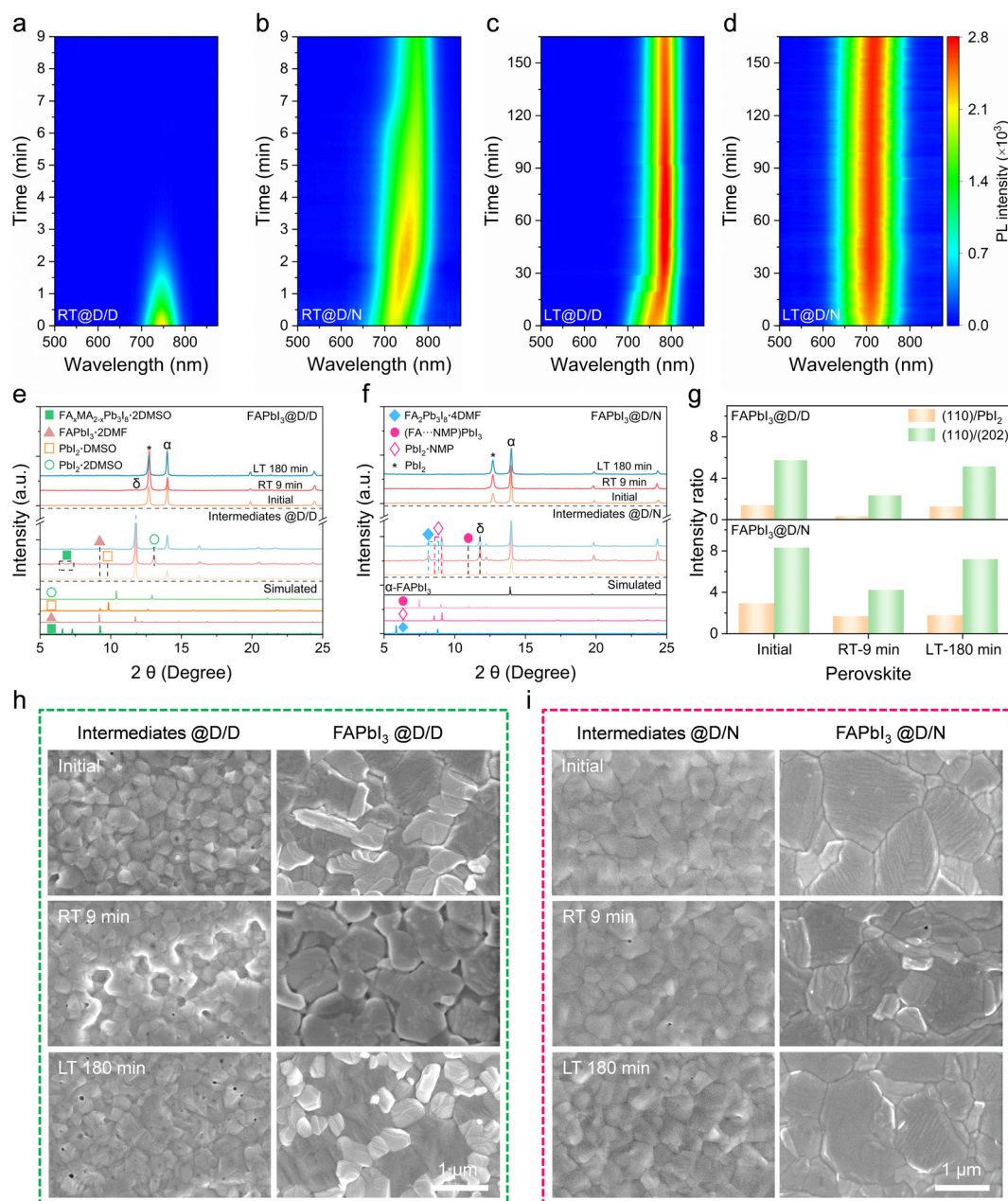


Fig. 3 Effect of LT treatment of intermediate phase films on perovskite nucleation and crystallization processes. (a)–(d) *In situ* PL intensity evolution of the intermediate@D/D and intermediate@D/N phases at room temperature (RT) and low temperature (LT). (e) and (f) The corresponding XRD patterns of the intermediate films at the initial state (0 min), after 9 min storage at room temperature (RT 9 min) and 180 min storage at 0 °C (LT 180 min), and the corresponding perovskite films after 150 °C annealing. (g) Peak intensity ratios of perovskite (110) to that of the Pbl₂ (001) and perovskite (202) peaks. (h)–(i) The top-view SEM images of the intermediates and corresponding perovskite films with different treatments.

After 60 min, the PL peak intensity showed a steady downward trend, due to the spontaneous decomposition of the intermediate phase even at LT and gradual formation of defects. The PL peak shifted to 785 nm after 120 min and became more symmetric in shape, suggesting that the conversion to δ -FAPbI₃ was suppressed. The PL peak intensity of the intermediate@D/N film also gradually increases and then remains unchanged (Fig. 3d), indicating that the intermediate@D/N phase is also more stable at LT. The PL peak of the intermediate@D/N also showed a slight redshift from 709 to 721 nm but did not shift further (Fig. S18b, ESI[†]), indicating that the intermediate phase with the solvent adduct is stable at this temperature. The intermediate can be rapidly transformed into the α -FAPbI₃ phase upon 150 °C annealing. The reason for the more stable intermediate at LT is probably due to slower solvent evaporation and suppressed decomposition of the Cl-containing intermediates.³⁶

The decomposition of the intermediates and formation of the δ -phase FAPbI₃ would affect the phase composition and crystallinity of the final perovskite film, as shown in the upper part of Fig. 3e and f. The ratio of the (110) peak of the perovskite (13.95°) to the (001) peak of PbI₂ (12.71°) is shown in Fig. 3g. A higher intensity ratio of the perovskite (110) and (202) peak (at 24.35°) indicates a good texture and crystallinity of the film.³⁷ The FAPbI₃ films derived from intermediate films stored at LT showed higher perovskite/PbI₂ ratio even with a processing delay time t_2 of 180 min. Fig. 3g shows the ratio of the (110)/(202) peaks of the perovskite phase. The perovskite films derived from the intermediate phase films stored at LT for 180 min showed a similar crystallinity and orientation as the initial film.

Fig. 3h and i show the top-view SEM images of the intermediate@D/D and intermediate@D/N films and their derived FAPbI₃ perovskite film after annealing at 150 °C. The morphology of the intermediate phase is different to the perovskite films, with smaller grain sizes. The change of intermediate@D/D at RT is most significant due to the volatilization of the solvent and MACl (or its byproducts). Changing the solvent to NMP or storing at a lower temperature can stabilize the intermediate. Only a few pin-holes can be seen from the RT-intermediate@D/N and LT-intermediate@D/D films. The LT-intermediate@D/N film is most stable and shows almost the same morphology as the initial film. The surface roughness analysis of the final perovskite films showed a similar trend. The perovskite films derived from RT-intermediate@D/D exhibit a higher roughness compared to the other counterparts (Fig. S19, ESI[†]).

Therefore, the evolution of the intermediate phase can be described by the proposed pathway shown in Fig. S20 (ESI[†]). For the D/D solution, a variety of different PbI₂ adducts are formed. When FAI/MACl is added, the initial intermediate@D/D intermediate phases consist of MA_xFA_(1-x)PbI_(3-x)Cl_x, PbI₂·2DMSO, FA_xMA_{2-x}Pb₃I₈·2DMSO, and FAPbI₃·2DMF. MACl, DMF and DMSO volatilize at RT, resulting in the decomposition of the MA_xFA_(1-x)PbI_(3-x)Cl_x intermediate, and eventually transformed into the δ -phase. When the MA_xFA_(1-x)PbI_(3-x)Cl_x

intermediate phase is stored at LT, the decomposition is suppressed to a large extent, and conversion to the α -phase is achieved, thus extending the t_2 window to 180 min. In the D/N solvent system, the MA_x(FA·NMP)_(1-x)PbI_(3-x)Cl_x phase was formed when FAI/MACl is added and is easily converted into the α -phase. Due to the strong hydrogen bond between NMP and FAI, MACl and NMP are less prone to volatilize, resulting in the formation of only a small amount of δ -phase with a long delay t_2 . The volatilization of MACl and NMP is further suppressed by the LT treatment, thus maintaining the intermediate phase structure of MA_x(FA·NMP)_(1-x)PbI_(3-x)Cl_x for a long time.

We first tested our strategy of stabilizing the perovskite intermediate in small area device fabrication. PSCs were fabricated with an n-i-p structure of ITO/SnO₂/perovskite/spiro-OMeTAD/Au. The PSCs fabricated with different t_1 (0 min and 60 min, Fig. S21, ESI[†]) showed similar efficiency, suggesting that t_1 is not an efficiency limiting factor for large-area PSC fabrication. Therefore, we turn our attention to t_2 , the process time of perovskite intermediates. Fig. 4a shows the J - V curves (reverse) of PSCs prepared with the different solvent systems at RT and LT. The RT-FAPbI₃@D/D device exhibited a PCE of 23.46% with a V_{OC} of 1.156 V, while the RT-FAPbI₃@D/N device displayed a higher PCE of 25.00%, with a V_{OC} of 1.197 V. The V_{OC} improvement suggests a higher quality FAPbI₃ film with the D/N solvent system.³⁸ The performance of the devices derived from the LT strategy also showed some improvement. The LT-FAPbI₃@D/N PSCs reach a PCE of 25.21% with low hysteresis (Table S3, ESI[†]). The stabilized PCE of the LT-FAPbI₃@D/N PSCs is 24.83%, measured at the maximum power point (Fig. S22, ESI[†]). Fig. S23 (ESI[†]) shows the external quantum efficiency (EQE) spectra of typical devices. The integrated J_{SC} values for the LT-FAPbI₃@D/D and LT-FAPbI₃@D/N devices were 24.40 mA cm⁻² and 24.97 mA cm⁻², respectively, consistent with the J_{SC} values extracted from the J - V curves.

The RT-FAPbI₃@D/N PSCs also possess satisfactory reproducibility with an average PCE of 23.20% (Fig. 4b and Fig. S24, ESI[†]), which was much higher than the value of 20.29% for RT-FAPbI₃@D/D. Using the LT strategy, the PSCs showed even better reproducibility. The average PCE of the LT-FAPbI₃@D/N devices was up to 23.80% at t_2 of 120 min or less (Fig. 4b and Fig. S25, ESI[†]). To show the difference in film uniformity, large-area PSCs with an effective area of 1.00 cm² were also fabricated (Fig. 4c and Table S4, ESI[†]). The PCE of the large-area RT-FAPbI₃@D/D device was 20.02%, and the PCE increases to 21.22% using the LT strategy. The PCE of the LT-FAPbI₃@D/N device was further increased to 23.60%. The stable power output (SPO) of the RT-FAPbI₃@D/N device reached 23.04% (Fig. S26, ESI[†]). We also measured the storage stabilities of the 1.00 cm² devices. The RT-FAPbI₃@D/N PSCs maintained approximately 95% of the initial efficiency after 2400 hours of storage in a dark nitrogen atmosphere (Fig. S27, ESI[†]).

The use of LT strategies is expected to prolong the process window due to the better stability of the intermediate phases at LT. Firstly, the J - V curves of small-area devices derived from intermediate phase films fabricated at RT and LT with different

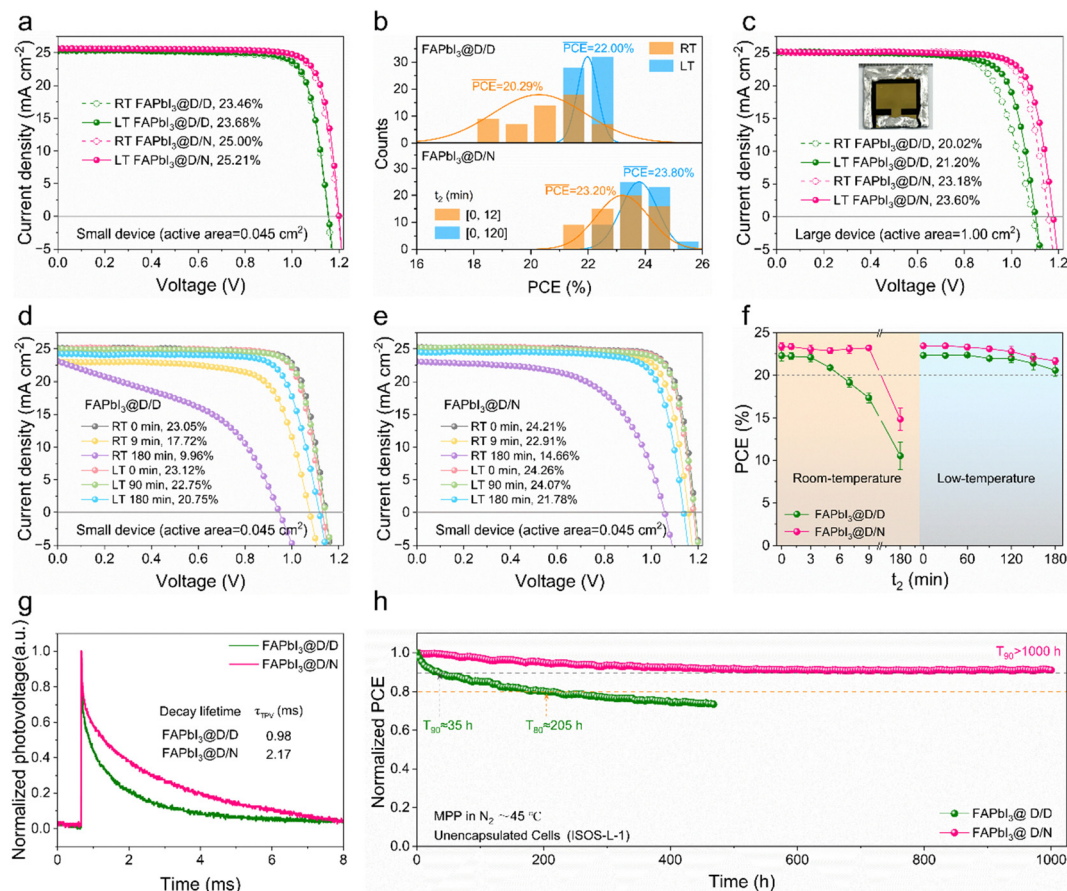


Fig. 4 Photovoltaic performance of PSCs based on perovskite films prepared with different solvent systems at RT and LT. (a) Typical $J-V$ curves of small-area devices (0.045 cm^2). (b) Histogram of the PCE value among 60 PSCs. (c) $J-V$ curves of 1.00 cm^2 devices. Comparison of the $J-V$ curves of perovskite intermediate-derived devices at different t_2 . (d) FAPbI₃@D/D and (e) FAPbI₃@D/N. (f) The PCE of perovskite intermediate phase-derived devices with different processing times of t_2 at RT and LT. (g) TPV carrier recombination lifetime of the device. (h) Unencapsulated PSC performance tracking at the MPP under continuous 1-sun equivalent white-light LED illumination for 1000 hours in a N_2 atmosphere.

t_2 values were compared to show the influence of temperature and t_2 on the PSC performance (Fig. 4d and e). The PCE generally decreased with increasing t_2 at RT. The PCE of the FAPbI₃@D/D devices showed significant PCE decay within 9 min and further decayed at 180 min (Fig. 4d and f). The FAPbI₃@D/N devices showed a more stable PCE in the initial 9 min, but still exhibited large decay at 180 min, with PCE decreasing from 24.21% to 14.66%. This was mainly because the prolonged storage at RT affected the morphology and composition of the intermediate film. The LT strategy can greatly prolong the process window. In contrast, the intermediate phase films were well preserved at LT. Remarkably, the PCE of LT-FAPbI₃@D/N PSCs remained close to 22%, even with a long t_2 of 180 min (Fig. 4e and f). This presents the longest process window (t_2) ever reported in the literature. In addition, we also studied the optoelectronic properties and device performance from a perovskite film derived from an intermediate phase at different storage temperatures and found that a storage temperature of 0°C is optimal for the film deposition (Fig. S28, ESI[†]). A storage temperature lower than 0°C reduces the PCE slightly and costs more energy for cooling. To strike a

balance between favourable device performance and energy expenditure, the storage temperature is set to the range of $0-5^\circ\text{C}$.

The carrier recombination properties of the perovskite films were a direct indication of the film quality. Steady and time resolved photoluminescence (PL) of the perovskite films were characterized to analyse the non-radiative carrier recombination. The FAPbI₃@D/N perovskite film exhibited enhanced PL intensity (Fig. S29a, ESI[†]), and longer PL lifetime (Fig. S29b and Table S5, ESI[†]). We also used space charge-limited current (SCLC) measurements to characterize the trap densities of these different films,³⁹ and the calculated defect densities of the FAPbI₃@D/D film and the FAPbI₃@D/N film were $6.89 \times 10^{15} \text{ cm}^{-3}$ and $5.51 \times 10^{15} \text{ cm}^{-3}$, respectively (Fig. S30, ESI[†]). To support this, leakage currents from the dark $J-V$ curves of PSCs were also examined (Fig. S31a, ESI[†]). The lower non-radiative recombination and defect density of the FAPbI₃@D/N films may be due to less impurity of the δ -phase and PbI₂ and less grain boundaries with large grain size in the film (Fig. S32b, ESI[†]). Carrier separation and extraction of the devices are also important to their photovoltaic performance.

The built-in potential (V_{bi}) of the PSCs provides a driving force for separating photogenerated carriers,⁴⁰ usually corresponding to the V_{OC} in the devices. The V_{bi} of PSCs was evaluated by Mott-Schottky analysis (Fig. S31b, ESI†). The FAPbI₃@D/N device exhibited a V_{bi} of 1.05 V, higher than the value of 0.93 V for the FAPbI₃@D/D device.

To gain more insight into the carrier transport and recombination kinetics at the device level, the charge recombination lifetime (τ_{rec}) and charge transport lifetime (τ_{tr}) of the PSCs were attained *via* transient photovoltage (TPV) and transient photocurrent (TPC) measurements.⁴¹ Compared with the FAPbI₃@D/D PSCs, the FAPbI₃@D/N PSCs exhibited a longer τ_{rec} (Fig. 4g), and a shorter τ_{tr} (Fig. S33, ESI†). This unambiguously demonstrates the improved carrier transport and suppressed carrier recombination in the FAPbI₃@D/N PSC at the device level. Therefore, the FAPbI₃@D/N perovskite films exhibited improved film quality and better carrier extraction, which led to substantial enhancement in the PSC efficiency and stability. We aged the unencapsulated devices under the conditions of 45 ± 5 °C with a white LED equivalent to 1 sun continuous light, and MPP tracking (Fig. 4h). The FAPbI₃@D/N device showed better long-term operational stability than the

FAPbI₃@D/D device. The FAPbI₃@D/N device maintained more than 91% of the initial efficiency after 1000 hours of MPP tracking under one sun illumination (white LED). The T_{80} of the FAPbI₃@D/D device was only 206 hours. This poor stability may be due to the excessive PbI₂ and defects in the perovskite film.⁴² To avoid the effect of hygroscopic additives in spiro-OMeTAD, we used undoped P3HT as the hole transport layer (HTL) to test the humidity stability of the PSC. As shown in Fig. S34 (ESI†), the unencapsulated LT-FAPbI₃@D/N and LT-FAPbI₃@D/D devices show better stability, maintaining 90.8% and 72.7% of the initial efficiency after aging for 216 h at 85% relative humidity, while the RT-FAPbI₃@D/D device only retains 57.5%.

We extended the LT-treatment to large-area films to demonstrate its capability in upscaling PSC fabrication. A large (6 cm × 6 cm) perovskite film (Fig. S35, ESI†) was prepared by the LT method with a long processing window ($t_2 = 120$ min). PL mapping showed uniform PL intensity across the film (Fig. S36a, ESI†). The UV-vis absorbance and PL lifetime at different locations on the diagonal line were almost identical (<5% variation, Fig. S36c and d, ESI†). The film was also cut into 16 small (1.5 cm × 1.5 cm) pieces to fabricate PSCs. The PCEs of

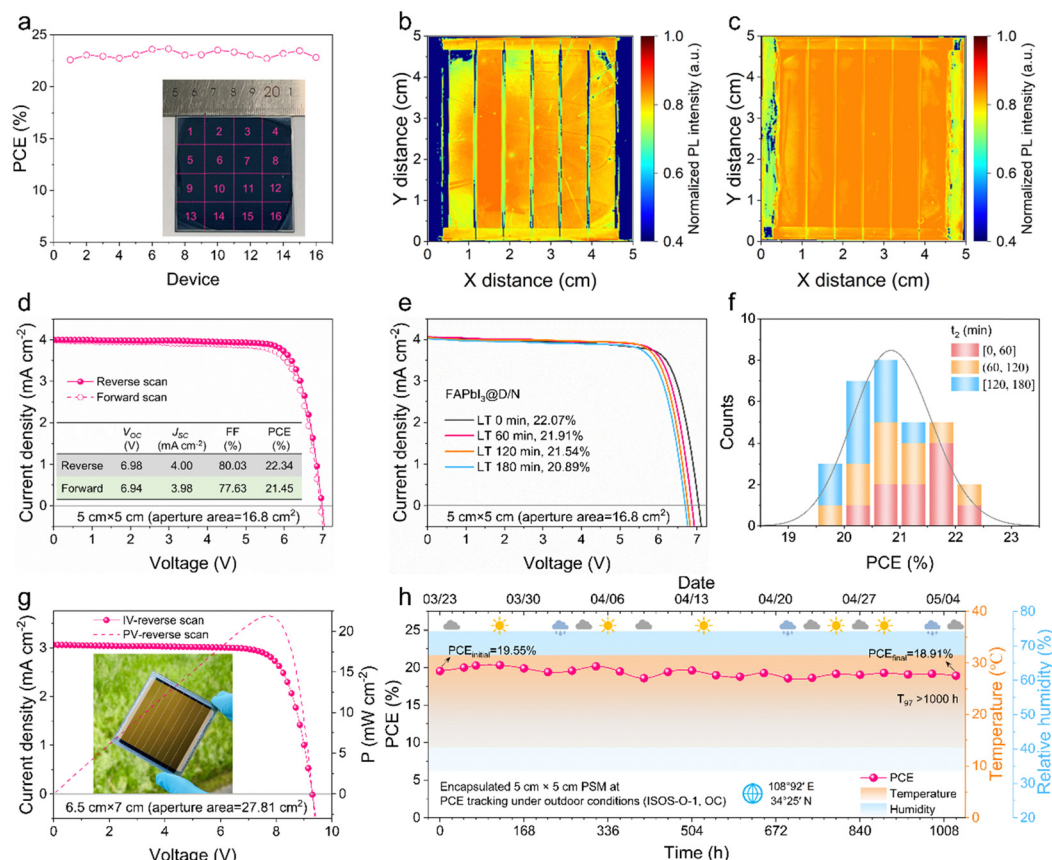


Fig. 5 Characterization of the photovoltaic performance of PSMs. (a) The PCE of devices derived from different regions of 6 cm × 6 cm FAPbI₃@D/N perovskite films. (b) and (c) PL mapping of the PSMs based on films of FAPbI₃@D/N stored at RT for 15 min and FAPbI₃@D/N stored at LT for 180 min. (d) J - V curves of the champion 5 cm × 5 cm PSM. (e) J - V curves of a 5 cm × 5 cm PSM derived from FAPbI₃@D/N with different t_2 at LT. (f) The statistics of PCE distribution for PSMs. (g) J - V curves (solid line) and power-voltage curves (dotted line) of the champion 6.5 cm × 7 cm PSM. (h) Outdoor stability of an encapsulated 5 cm × 5 cm PSM exposed to outdoor conditions and held at open-circuit conditions.

the 16 individual devices were consistent at approximately 23% (Fig. 5a and Fig. S37, ESI[†]).

Furthermore, 5 cm × 5 cm and 6.5 cm × 7 cm PSMs with 6 or 8 sub-cells connected in series were prepared with P1–P2–P3 laser scribing. Details of the PSM fabrication process are shown in Fig. S38 (ESI[†]). PL mapping was employed to examine the uniformity of the 5 cm × 5 cm PSMs derived from different intermediate phases. The PSMs derived from FAPbI₃@D/N with a t_2 of 15 min at RT showed a low PL intensity and nonuniform distribution (Fig. 5b), indicating that the perovskite films were of poor quality and had many macroscale defects. In contrast, the PSMs prepared with a t_2 of 180 min at LT still showed a uniform PL intensity (Fig. 5c), which can be attributed to stabilization of the intermediate phase structure at LT. The LT-treated PSM also showed higher PCE due to the higher film quality and uniformity (Fig. S39, ESI[†]). Fig. 5d shows the J – V curve of the 5 cm × 5 cm PSM with an aperture area PCE of 22.34% and a fill factor (FF) exceeding 80%. The SPO of the PSM reached 21.58% (Fig. S40, ESI[†]). A comparison of the reported PSMs in the literature is summarized in Fig. S41 (ESI[†]).

Impressively, the intermediate@D/N films were stored at LT with t_2 increased from 0 to 180 min and showed no change in optical images (Fig. S42, ESI[†]). Fig. 5e shows the J – V curves of the PSMs with different t_2 at LT. A decent PCE of 20.89% was still maintained with t_2 of 180 min. The average PCE of the PSMs fabricated with $t_2 = 0$ –180 min was 20.88% (30 PSMs with $t_2 = 0$ –180 min were prepared, and the distribution of the performance parameters was statistically analysed; see Table S6, ESI[†]). Our results show that the LT treatment of the perovskite intermediate phase greatly extends the processing window for fabricating a PSM at the laboratory scale, which is helpful for further commercialization of PSCs.

By upgrading the laser scribing technique, the corresponding 6.5 cm × 7 cm module demonstrated a high efficiency of 21.91% with an aperture area of 27.81 cm² and a geometric fill factor (GFF) of 97% (Fig. S43, ESI[†]), corresponding to an active area efficiency of 22.59% (Fig. 5g). The certified best PCE of the PSMs reached 21.51% with an aperture area of 27.21 cm² according to J – V scanning (Fig. S45, ESI[†]). The SPO measurement method gave a stabilized PCE of 21.34% (Fig. S46, ESI[†]).

We have also conducted light soaking, high humidity and outdoor stability tests on the 5 cm × 5 cm PSMs. They showed excellent performance under continuous illumination (Fig. S47 and S48a, ESI[†]), with T_{80} reaching 550 hours in air. Moreover, the unencapsulated LT-FAPbI₃@D/N module can maintain more than 84% of the initial PCE after aging for 216 h at 85% relative humidity (Fig. S48b, ESI[†]). We evaluated the outdoor stability of the encapsulated PSMs under ambient conditions based on the International Summit on Organic Photovoltaic Stability (ISOS) protocols.⁴³ An outdoor test platform designed in-house was used for the stability measurement (Fig. S49, ESI[†]), and the temperature and humidity of the outdoor environment were recorded in real time. Fig. 5h shows the actual PCE of an encapsulated 5 cm × 5 cm FAPbI₃ PSM under outdoor conditions over time. After 1032 hours of

continuous exposure under outdoor ambient conditions, the module still showed a slight PCE decay from 19.55% to 18.91% (Fig. S50, ESI[†]). The outdoor T_{80} lifetime of the module is predicted to be 6300 hours through linear extension, which provides a basis for future outdoor applications of PSMs.

Conclusion

In summary, we report a strategy for stabilizing the perovskite intermediate phase to expand the processing window, enabling the fabrication of large-area perovskite thin films. The interaction between the solvent and perovskite precursor was enhanced to form a stable intermediate phase with a stronger binding solvent of NMP. The α -phase was effectively formed even with an annealing delay of 9 min at RT. In addition, by storing the perovskite intermediate phase at LT, loss of solvent and decomposition of the intermediate phase was suppressed, and the annealing delay time window was further increased to more than 180 min. Using the solvent engineering and LT treatment strategy, large-scale deposition of high-quality large-area FA-based perovskite films was realized. PCEs of 25.21% for small-area devices (0.045 cm²) and 23.60% for large-area devices (1.00 cm²) were obtained. We further demonstrated this strategy with the fabrication of PSMs that achieved state-of-the-art champion efficiencies of 22.34% and 21.91% (certified efficiency of 21.51%, stabilized at 21.34%) with aperture areas of 16.8 cm² and 27.81 cm², respectively. More importantly, even with a long annealing delay time of 180 min, the modules still achieved a PCE of 20.89% with good repeatability, achieving the longest processing window for the two-step deposition of perovskite. Our strategy of stabilizing the perovskite intermediate phase with an ultra long processing window will facilitate up-scaling deposition of perovskite films at greater substrate area and bigger batches, paving the way towards high-performance commercial PSMs.

Conflicts of interest

There are no conflicts to declare.

Acknowledgements

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